9 Brain-Machine Interfaces

Predicting is more advantageous than simply reacting.
—Simone Schütz-Bosbach and Wolfgang Prinz [SBP07]

9.1 Overview

No aspect of neural engineering has captured the popular imagination as much as brain machine interfaces. From the medical grail of producing thought-powered prosthetics for the disabled, the military grail of thought-powered ordinance, to the entertainment grail of thought-powered fantasy games, brain machine interfaces are on the minds of many people.

Our advances in brain-machine interfaces have been almost entirely through the use of empirical *black box* statistical models to solve the decoding problem between neuronal recordings and motor movements. Some of the world's best minds in neuroscience and statistical theory have produced a set of powerful solutions to this *decoding problem*. Yet the results so far could have been produced without knowing anything much about the brain that produces the signals beneath our electrodes.

The best of the generalized statistical models employed for such decoding problems observes the brain activity, trains a model by fitting data, and uses that model fit to decode future data. No one has yet, to the author's knowledge, produced fundamental and biophysically based models that can observe and track such brain circuitry *without data fitting*, and create recursive predictive models. Are we at the point where our computational neuroscience of, say, the motor system, is sufficiently advanced that we can begin to shed some light inside the black box? Perhaps soon. Let's examine this intriguing field.

9.2 The Brain

We begin our tour of brain-machine interfaces by introducing, in figure 9.1, a schematic of the brain. The motor and sensory sheets of the brain have highly localized functions. By this we mean that if you stimulate while following along their length, you can get motor

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

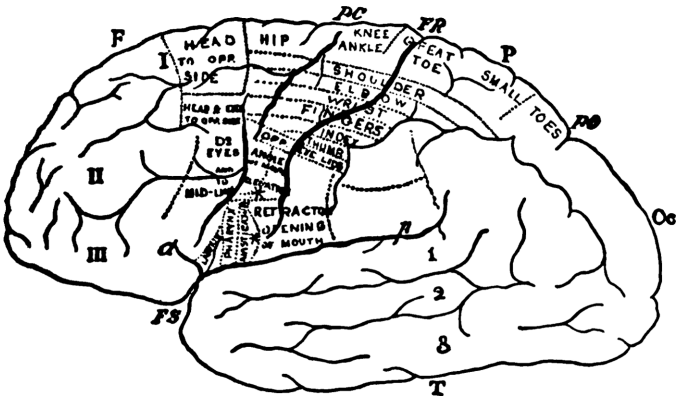


Figure 9.1
Schematic of topological layout of the motor *homunculus* of the brain. The figure represents William Gowers's mapping [Gow01] onto the human brain of Victor Horseley's findings from the stimulation of macaque [BH90a] and orangutan [BH90b] primates. Reproduced from [Gow01].

responses, or sensory responses, respectively, that reflect a 1:1 spatial mapping of the body. Figure 9.1 shows that this mapping is distorted with respect to body size, and we call this small humanoid map the homunculus. If one were to map tones in the auditory area, you would get a tonotopic map of frequencies. If you map the visual field in the primary visual cortex, you would get a spatial map of the visual field. It is the primary motor and sensory regions of the brain that have been most carefully recorded from, in an effort to construct brain-machine interfaces.

9.3 In the Beginning

The study of brain-machine interfaces really begins in 1982 with the seminal work of Apostolos Georgopoulos [GKCM82]. In these experiments, Georgopoulos studied neurons recorded from the primary motor cortex of the brain, and recorded single neurons while a monkey played a game with a manipulandum. The monkey's task was to first center the manipulandum over a light in the center of the board. Once held there, the task was to next move the device over one of eight possible lights arranged at 45-degree angles in a circular field. If this was done correctly, the monkey received a juice reward. An example of the trajectories of an experiment are shown in figure 9.2C.

The response of a single cell during trajectories in eight directions are shown in figure 9.3. There is clearly a strong directional preference in the firing rates associated with movement in preferred directions, and these firings increased prior to movement onset. Interestingly, in directions away from the preferred one, firing would decrease and at times stop at movement onset and in the period immediately afterwards. Georgopoulos fit these firing rates to a cosine tuning function:

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

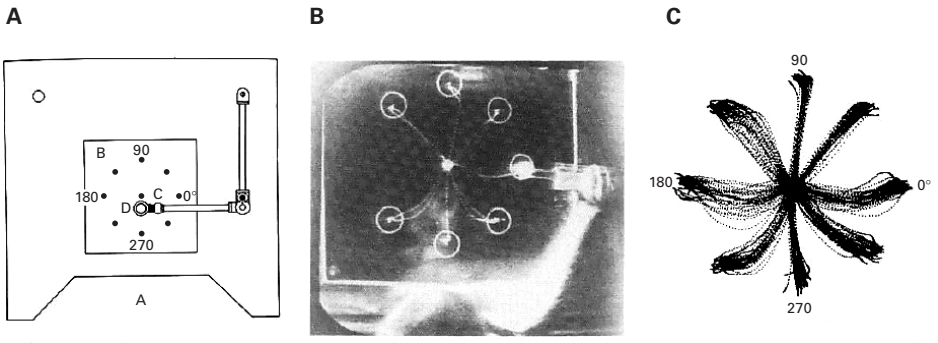


Figure 9.2
Schematic of experiment (A), picture of the monkey performing an experiment (B), and trajectories overlapped at the end of an experiment (C). Reproduced from [GKCM82] with permission.

$$z_k = h_0 + h_p \cos(\theta_k - \theta_p)$$

The firing rate of a given neuron, z_k, was modeled as a baseline rate, h_0, plus a cosine function of the difference between the movement direction, θ_k, and a preferred direction for that neuron, θ_p (with an adjustable constant h_p). From the trigonometric identity that none of us could remember as school children,

$$cos(a - b) = cos(a)cos(b) + sin(a)sin(b)$$

one gets

$$z_k = h_0 + h_1 sin(\theta_k) + h_2 cos(\theta_k)$$

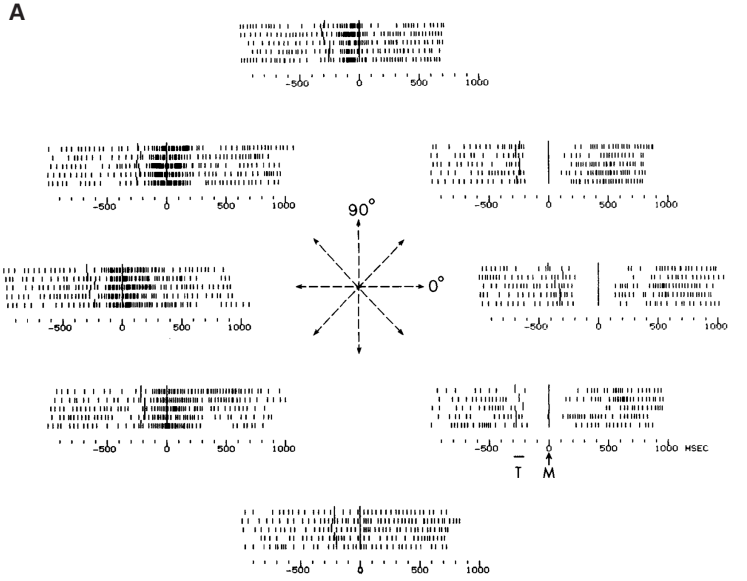
because the preferred direction, θ_p, for each neuron is a constant and can be lumped into the constants h_1 and h_2.[1]

Georgopoulos found that when a given movement in a particular direction was made, most of the cells that fired had directional preferences close to the direction of movement. This work suggested that were one to have recorded from these cells at the same time, a sum of the vector angles of the cells that fired would predict the angle of the arm movement. *This finding is the foundation of all brain-machine interfaces that we presently have, and are about to discuss.*

In 1986, Georgopoulos published the three-dimensional version of this experiment [GSK86]. Again, summing the population vectors for multiple cells was highly predictive

1. Be very careful in assuming that because the cosine function fits these data, that the neurons are little cosine generators. In further work from Georgopoulos and colleagues [AG00], they explored a variety of uni- and multimodal functions. A generalized form of circular distributions, the von Mises function, can often fit these tuning curves better, and represents a sum of Bessel's functions. The reader should think of cosines as a convenient way to represent a unimodal continuous peaked function in representing a neuron's preferred direction.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026



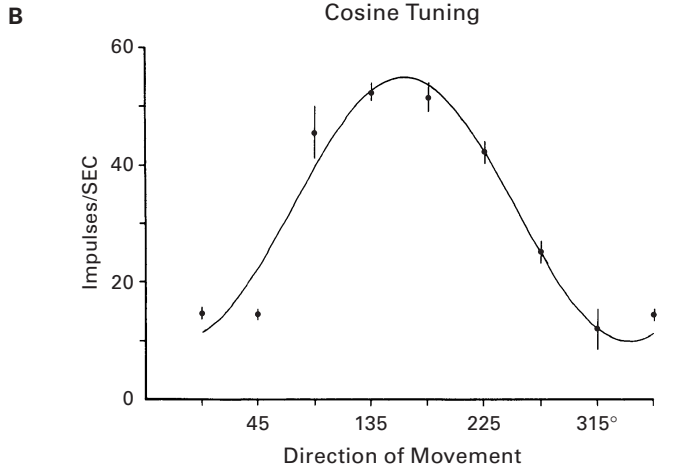


Figure 9.3
Raster plots showing the sequential firing times of spikes from a single cortical cell for different directional movements in A, and the fit of these firing rates to a cosine tuning curve in B. Reproduced from [GKCM82] with permission.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

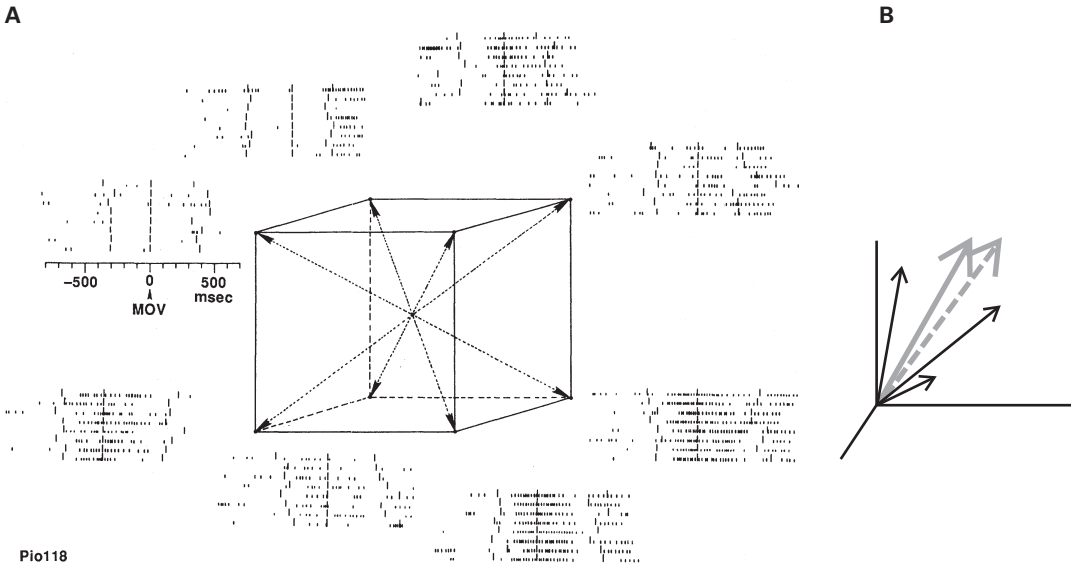


Figure 9.4
Three-dimensional version of the arm trajectory experiment. A single cell firing in response to direction in A, and a three-dimensional plot illustrating how individual cell cosine tuning vectors (black arrows) can be summed through the population model to reflect a direction in three-dimensional space (gray solid arrow) that closely approximates the true movement direction (gray dashed arrow). Image in A is reproduced from [GSK86] with permission.

of the actual trajectory that the arm made within the eight possible vertices of the cube, as shown in figure 9.4.

9.4 After the Beginning

Within several years of Georgopoulos's work, our technology advanced to the point that placing arrays of microelectrodes offered the possibility of recording from more than one cell at a time, and extracting their spike times in real time.

In 2002, John Donoghue's group published a paper that examined less constrained movements than we showed above, and with simultaneous recordings from multiple neurons [SHP+02]. In these experiments, a target appeared at multiple positions on the screen, and a monkey needed to trace the trajectory using a manipulandum. Figure 9.5 shows a combined trajectory of the hand movement (green) as well as the brain-firing reconstruction (blue).

In contrast to the population vector approach, Serruya and colleagues [SHP+02] created a linear filter that converted neuronal firing to hand position. They established a firing rate matrix for k neurons, n, each observed over fifty time bins as

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

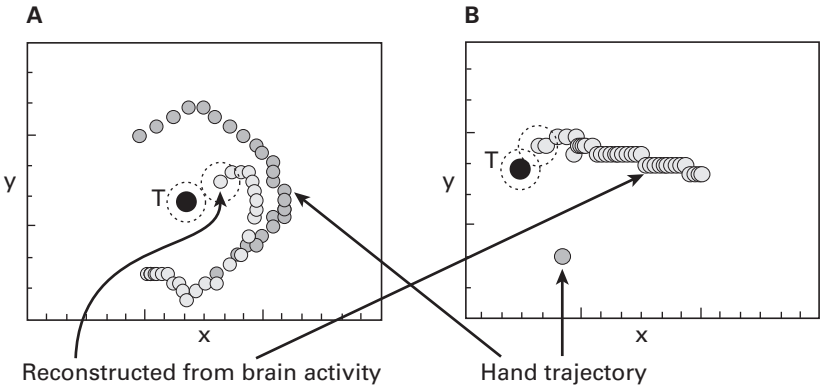


Figure 9.5
Target, T, hand trajectory, and reconstruction of hand trajectory from neuronal activity given filter $\hat{f}$. A shows the hand and reconstructed trajectory. B shows a trajectory where the monkey did not move its hand. Reproduced from [SHP+02] with permission.

$$R = \text{firing rates} = \begin{pmatrix} n_1^{t-49} & \cdots & n_k^{t-49} \\ \vdots & \ddots & \vdots \\ n_1^{t} & \cdots & n_k^{t} \end{pmatrix}$$

where

$$Rf = x$$

and x is the hand position. The filter, f, converts firing rates to position. Here, hand position is modeled as a linear combination of firing rates. The *best* filter for this, $\hat{f}$, is the solution of an equation that we solved in chapter 1, with equations (1.28). It is an over-determined problem, which we solve with least squares

$$R^T R \hat{f} = R^T x$$

$$\hat{f} = (R^T R)^{-1} R^T x$$

$$R\hat{f} = R(R^T R)^{-1} R^T x$$

Schematically, these equations are shown in figure 9.6. Each row of the matrix R, representing a collection of neurons $1 \ldots k$, forms an inner product with each column of the filter, $\hat{f}$, producing two numbers: a pair of coordinates x and y. Least squares finds the best filter given the data to accomplish this task. This simple linear filter solves a stationary system.[2]

2. A Wiener filter solves for the optimal filter for a stationary system where the information being destroyed by noise or damping is exactly balanced by the incoming information such as our spikes [Gel74]. You will see Wiener filters being frequently applied to such linear problems, but rarely will the underlying requirements for the existence of the filter be fulfilled.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

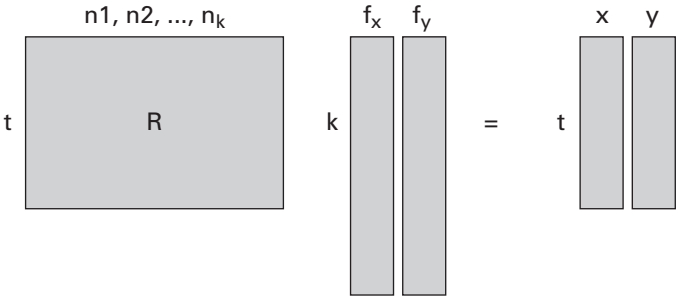


Figure 9.6
Schematic of linear filtering of spike times.

Serruya and colleagues then made a peculiar observation. Sometimes the monkey didn't move its hand, but nonetheless was able to rapidly move the cursor to the target using brain activity alone (figure 9.6).[3] It's the serendipitous findings that are so often the most important.

Two years later, this same group demonstrated that both hand velocity and position could be independently encoded or decoded from these same experiments [PFHD04]. There is something very subtle here. In Georgeopoulos's experiments, the task was highly constrained. The monkey moved along prescribed trajectories that were really quite different from each other and algorithms that attempted to extract these different directions could be very successful. In the pursuit task, where the directions were part of a random walk, the situation is far more complex. Adding more features to the model for which qualitative features in hand movement are encoded, can give improved performance of such a filter. But how many features can be extracted from neurons firing in the motor cortex? The question is a deep one.

Is the motor cortex thinking? It is probably not thinking nearly as much as it is serving as a final common pathway [She06] for decisions made elsewhere in the brain. Of course, I don't know that this is true. One does suspect from research in recent years that whatever projects to the motor cortex may well read its output activity in order to form a feedback loop with it. Such efference copies have long been considered in motor movement. They might be very useful to a part of the brain attempting to drive the motor cortex using predictive coding [Bal97]; but they also are very convenient for people trying to interact with the brain using interfaces in the twenty-first century. The point that I want to emphasize is that just because one can find representation of particular features within the primary motor cortex does not necessarily imply that those neurons are using those properties themselves. But it is nice for us that these representations are there.

3. It turns out that giving the brain visual feedback, closing the loop in contrast to the open-loop experiments of Georgeopoulos, was critical to have observed this (we will return to this issue in section 9.7)

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

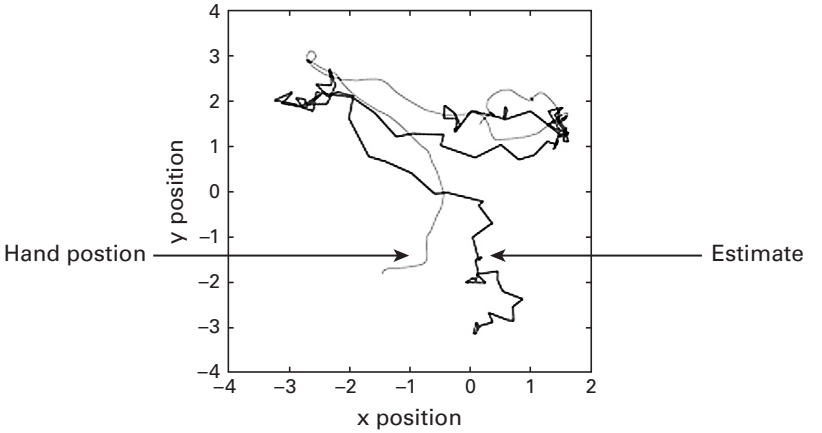


Figure 9.7
Two-dimensional hand position and estimate. Reproduced from [PFHD04] with permission.

Using position and velocity, Paninski and colleagues [PFHD04] demonstrated that they could improve their reconstruction of hand position using the data from Serruya et al. [SHP+02]. They had an interesting approach—assuming that hand kinematics were input random variables of position and velocity, and that neural activity was the response. Their reconstruction of trajectory, as shown in figure 9.7, is compelling in interesting ways. One cannot help but keep asking oneself, on inspection of such reconstructions, what qualities are "good," and which are not. It is clear from this undoubtedly exemplary example that the correlation between the reconstruction and reality is better than the actual error in position. Recall from chapter 8 that the model and reconstruction are here shown in a different space from reality, which is much more complex. Examine the geometry of model inadequacy in figures 8.2 and 8.4. It is interesting to speculate how adapting some of the model inadequacy formalism from chapter 8 might improve some of the position errors in figure 9.7. But speculation on decreasing position errors in [PFHD04] is not the reason to introduce this remarkable work.

One of the most remarkable aspects of [PFHD04] was that by randomly picking a small subset of the recorded neurons, the researchers achieved incredibly high reconstruction accuracy, as shown in figure 9.8.

This is very different from the Georgopoulos result where only the neurons that gave significant direction tuning were considered in the analysis. Here, using a linear filter of position and direction, any sampling of twelve random cells gave excellent reconstruction accuracy. Were the investigators lucky in positioning their electrodes near the cells? Not at all. Such results imply that very high percentages of the neurons recorded within the region of the primary motor cortex have representations of both the position and velocity of the hand. What this means is that you can go into, say, the hand region of the motor cortex,

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

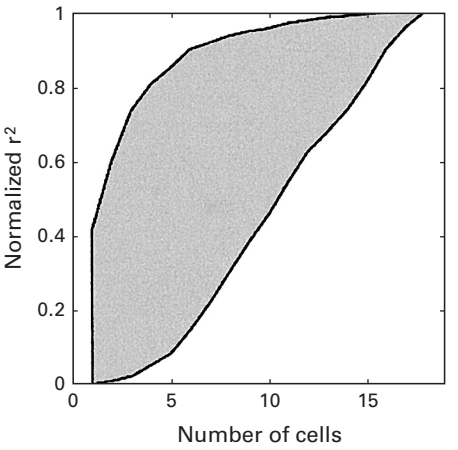


Figure 9.8
Range of reconstruction accuracy versus number of randomly selected neurons. Reproduced from [PFHD04] with permission.

grab a dozen or so neurons—it does *not* matter which ones—and reproduce the results of [SHP+02] shown in figure 9.6.

But why are such higher-order qualities of movement represented in Sherrington's *final common path* for motor movement? We think that at some point, the thinking is done, and motor systems are carrying out instructions generated upstream. Perhaps the motor instructions are more complex than just pulling levers of muscles (in primates, much of the output from primary motor cortex is flowing to spinal cord interneurons for further processing rather than directly to motoneurons). But also it may well be that a degree of feedforward and feedback flow of information to and from motor cortex is essential in this process. In [PFHD04] it was apparent that, for velocity tuning, there was a clearly predictive element emerging about 100 msec prior to motor action. One can look for higher-order derivatives of position as well (acceleration, jerk, snap, crackle, etc.), and find a (decreasing) representation of these quantities in such motor cortex neurons (see [WGB+06]).

There seems to be three possibilities for these representations: (1) they are just there for the convenience of twenty-first–century scientists wishing to construct brain-machine interfaces; (2) they are just there and the nervous system wastes energy transporting this information because it is too expensive to remove (the informational appendix of the motor system); or (3) they serve a purpose other than direct transmission of motor movement. As in other paradigms of anticipation found in nervous systems, whether the stimulus expectancy of primates [RGDA97], or the omitted stimulus response of ants and bees [RG05], it is hard to write this book without thinking speculatively on the structure of the nervous system's possible predictor-corrector structures.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

In 2006, using the same linear filter for position from [SHP+02], the Donahue group showed that their original monkey experiments taught them how to translate this paradigm to humans [HSF+06]. They first demonstrated this in a patient with a high spinal cord injury. He was unable to move his arms or legs, and required a ventilator to breathe. They used the same microarray used in the [SHP+02] experiments, and implanted it in the small bump in the motor cortex affectionately known as the *hand knob* by those who map such things (figure 9.9C). An important difference between training such disabled people and neurologically intact monkeys is that the patient cannot be trained by moving his or her hands. In figure 9.9D a technician is seen providing cursor training early on in such tasks.

By watching the technician and *mirroring* in his mind, he learns to perform these tasks with his mind. He can often perform these tasks while chatting with the technician about tangentially related things. A rich theory in psychology helps us weave a network of brain mechanisms that underly such phenomena. The discovery of *mirror neurons* by Rizzolatti and colleagues [dPFF+92] has opened up a vista originally speculated on by William James [Jam90]. A century later, we have divided our nomenclature into instantly predictive coding of sensory phenomena by motor actions through generative models [KFF07], and future prospective coding of sensory events anticipated in a future time [SBP07]. Perhaps James's speculations filtered through mirror neurons and their implications render the observations of [HSF+06] plausible but no less remarkable. We will return to the broader implications of mirror neurons in chapter 13.

The relative ease of training and the ability to perform tasks with modest degrees of attention are rather different from the extracranial experience with brain-machine interface implementations [WMVS03]. Nevertheless, with sufficient training, scalp-based electroencephalograms (EEGs) can perform accurately once the subject (and algorithm) masters the learning curve [WM04]. Recent work with subdural sensing electrodes suggests that an intermediate location between inside the cortex and outside the head may provide an accurate signal for decoding without transgressing the cortex itself [SKM+07, MSF+10].

9.5 Beyond Bins—Moving from Rates to Points in Time

A brain-machine interface can respond only as fast as the signals being processed can be calculated.[4] In both [SHP+02] and [HSF+06], a 50-msec time bin was used to count spikes and form an average rate. We now look at efforts to minimize this by examining the timing of individual spikes themselves.

We begin with a rodent experiment. O'Keefe and Dostrovsky noted that, in addition to other complex sensory and arousal states that caused firing in hippocampal neurons, certain

4. We leave out for the moment the significant time lags introduced by the human brain between the presentation of a visual cue and the perception of such an event by the brain, which can typically take more than 100 msec.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

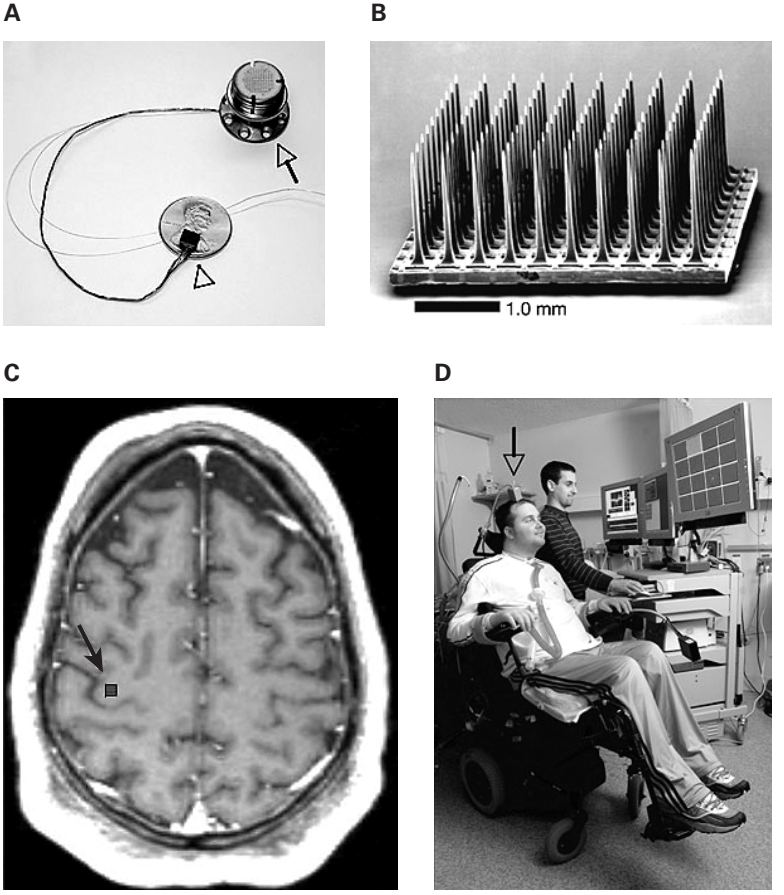


Figure 9.9
The microfabricated Utah array is shown in A and B; the positioning of the array on the *hand knob* of the motor cortex is shown in C; and the training of a quadriplegic subject with this implant with a technician is shown D. Reproduced from [HSF+06] with permission.

neurons were particularly sensitive to spatial orientation of the animal and position in space [OD71]. Later, it was observed that the firing of place neurons often occurs in preferred phases of what is called *theta* rhythm (generally 4 to 12 Hz in animal EEGs).[5] In fact, the phase that such place cells fire often *precessed* in relation to the phase of theta rhythm, as the animal entered the place-receptive field and moved through it [OR93].

5. A source of continual annoyance to clinicians is that human EEG "theta" rhythm is rather rigidly defined as 4 to 7 Hz. But brains are not standard clocks, and a more functional banding of neural frequencies, especially in animal behavioral studies, is often used.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

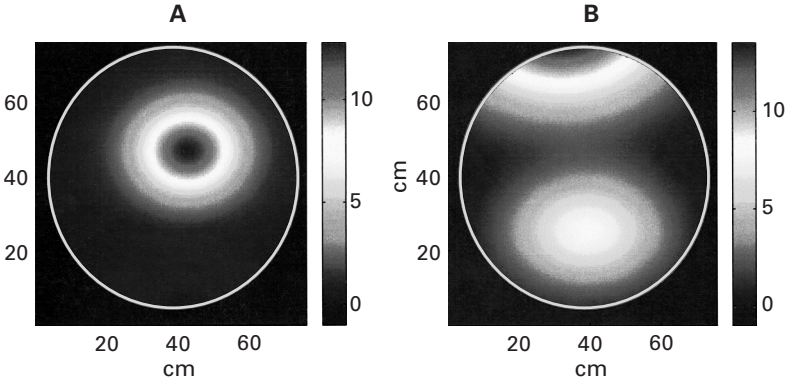


Figure 9.10
Inhomogenous Poisson model fit to firing of individual place cells within an open field arena. The gray scale is spikes per second, as indicated in the intensity bars. A shows a unitary receptive field, and B shows a split receptive field with two distinct regions where the place cell firing peaks. Reproduced from [BFT+98] with permission.

A Poisson process is a single-parameter random process in which the rate parameter, λ, is the expected number of events within a unit of time, and the probability of seeing a given number of events, k, within a unit length of time is proportional to $e^{-\lambda}(\lambda)^k/k!$. If the rate λ is itself time or position dependent, then the Poisson process is *inhomogeneous*.

Can one encode the continuous position of the animal, x, and construct a conditional probability for neural spiking, z, as $p = (z|x)$?

In 1998, Emery Brown and colleagues [BFT+98] used an inhomogenous Poisson process to fit to place cell firing and theta phase data as shown in figure 9.10. These maps represent the initial *encoding* process.

One then takes this encoding and uses it to observe the neural spiking and, in a Bayesian framework, *decodes* the posterior probability of position $p(x|z)$. [BFT+98] solved the problem of taking a point process of spike events and estimating the continuous position of the animal. The original Kalman filter discussed in chapter 2 estimated continuous variables from measuring continuous observables, so some adaptation was required.

Figure 9.11 compares a linear model and the estimates with the Bayes filter. In the Bayes filter, the underlying model is that of a random walk. This *generative model* has the effect of placing a huge prior constraint on each next estimate of the animal's position. Rats don't fly. The linear model contrasted in figure 9.11 had no such constraint.

One might then ask whether the simpler linear model might have done much better using a constraining generative model. Perhaps a simple linear Kalman filter, using spike rates instead of point processes, might perform admirably well given kinematic constraints and dynamics.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

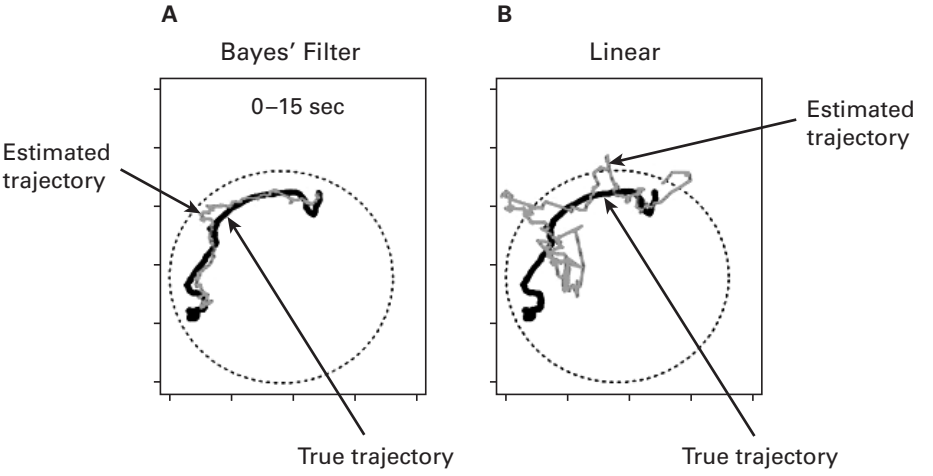


Figure 9.11
True and estimated trajectories using Bayes' filter (A) and a linear filter (B). Reproduced from [BFT+98] with permission.

9.6 Back from the Future

In [WGB+06] the authors took the data analyzed in [SHP+02] and [PFHD04]. They reanalyzed it using kinematic constraints and a generative model of limb dynamics, and used continuous measures of spike rates rather than point process observations of spike times. And they applied the linear Kalman filter as we have described in chapter 2.

They examined two tasks: a *pursuit* tracking task where the target moves in a random walk, and a *pinball* game where the target appears at random locations and the goal is to hit it. The kinematic variables will be position, as well as derivatives of position (velocity, acceleration, jerk, snap, crackle, etc.).

Linear models of hand kinematics include

$$z_k = h_0 + h_x v_{x,k} + h_y v_{y,k}$$

where z_k is the firing rate of a cell at time k, and the h's are linear functions relating the velocities v_x and v_y at time k to the neuron firing. Position would be linearly encoded as

$$z_k = f_0 + f_x x_k + f_y y_k$$

where the linear functions f relate position at time k to firing rate from the coordinates of hand position (x,y). [WGB+06] will also include higher-order derivatives in such linear functions (especially acceleration).

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

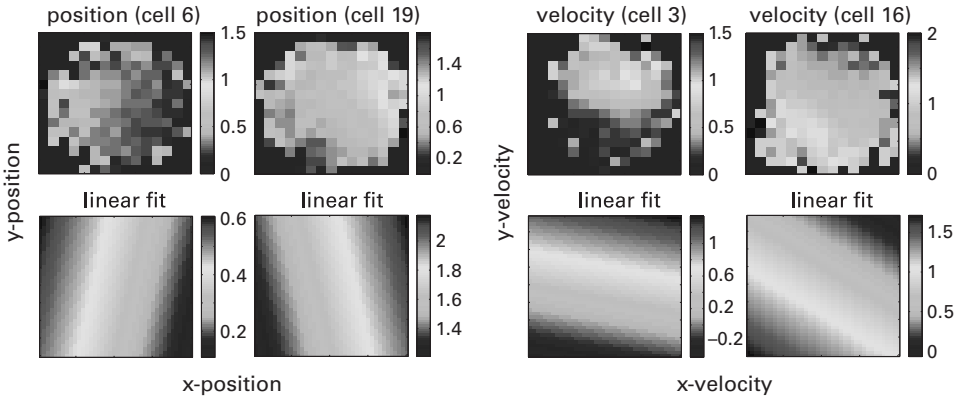


Figure 9.12
Cell firing frequency versus position or velocity as a function of position, with gray scale calibrated as shown for firing frequency. Below are the best linear fits. Reproduced from [WGB+06] with permission.

Figure 9.12 shows examples of kinematic data that was encoded into their linear functions for pursuit tracking. Hand position is plotted against the intensity map of cell firing frequency.

The first thing to discuss about figure 9.12 is that fitting linear models to receptive fields that are circular is problematic. Kalman filtering works when the models of the dynamics are better than random guessing. The authors describe these fits as *crude but reasonable* [WGB+06]. One might have said that the models are *not quite entirely unlike* the data.[6] Clearly, these fits can be improved with radial functions, and we will return to this issue later in this chapter. But for now, the point is that these linear fits capture *something* from these data that is useful, and the power of the Kalman filter is that it optimizes the uncertainty of this inadequate model.

Are these single neurons? [WGB+06] actually used any waveform that crossed a given threshold. So their firing rates undoubtedly included the possibility of multiple single neurons, and combinations of neurons. Isolating single neurons would require spike sorting, and matching multiple waveform features of individual spikes recorded at high sampling frequencies (see, for example, [SDW02]). The literature on *single-cell* spike rates and timings from electrode arrays is full of examples where a critical view of the quality of single-cell isolation can be raised. In [WGB+06] this issue is acknowledged. As for the model fits discussed earlier, whatever these signals are, single versus multiunit recordings, they *are* sufficient to permit hand kinematic estimation. On the other hand, this issue raises the important question of how detailed the data need to be to function in a brain-machine interface. Might local field potentials or local EEGs suffice? Do we need to use microfabricated electrode arrays that corrode over merely months in typical applications, rendering

6. A paraphrase of Douglas Adam's expression from his *Hitchiker's Guide to the Galaxy* [Ada79].

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

them useless in their present instantiations beyond a year or two? Might we be able to use recordings from the subdural or epidural space to achieve adequate brain-machine interface accuracy [SKM+07, MSF+10]?

Also important to note is the sparseness of neurons recorded by the arrays used in [SHP+02], [PFHD04], and [WGB+06]. There should be an order of magnitude more neurons from considerations of neuronal packing density, or from the contrast between extracellular and intracellular recordings. Shoham and colleagues [SOS06] call this the *dark matter* problem in the mammalian brain. Most of the signal recorded by microelectrode arrays may not be neuron firing. Most neurons may be quietly listening most of the time, rather than firing. We have been taught to be spike-centric—the mammalian cortex may not be.

With all of these cautionary issues in mind, let us assume an augmented state of position, velocity, and acceleration:

$$\mathrm{x}_k = \begin{bmatrix} x \\ y \\ v_x \\ v_y \\ a_x \\ a_y \end{bmatrix}$$

where k is an index of time such that $k = t/\Delta t$. Δt is the time over which the spikes are binned to derive the firing rates, 50 to 75 msec. Assume that the observations are firing rates z, such that for C neurons at time index k

$$\mathbf{z}_k = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_C \end{bmatrix}$$

and Z_k represents the history of measurements

$$\mathbf{Z}_k = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_k]^T$$

Next assume that the a posteriori probability of system state x_k, conditioned on measurments z_k, is

$$\mathrm{p}(\mathbf{x}_k|\mathbf{z}_k)$$

and now assume two Markov assumptions

$$p(x_k|x_{k-1}, x_{k-1}, ..., x_1) = p(x_k|x_{k-1})$$

where the hand kinematics really depend on only the most recent values of position, velocity, and acceleration, and

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

$$p(z_k|x_k, z_{k-1}) = p(z_k|x_k)$$

which tells you that neurons are forgetful of their past spikes, so you can forget them also in your formula relating neuronal spiking to hand kinematics. For now, assume that this relationship is instantaneous—we will change this ridiculous assumption shortly.

Now recall Bayes's rule from chapter 1:

$$P(x, z) = \text{Joint Probability} = P(x|z)P(z) = P(z|x)P(x)$$

which leads us to the central formulation in [WGB+06] and related literature

$$\mathrm{p}(x_k|\mathbf{z}_k) = \kappa \;\; p(z_k|x_k) \int p(x_k|x_{k-1})p(x_{k-1}|z_{k-1})dx_{k-1}$$

This is nothing more than Bayes' rule. Let's dissect it. $p(x_k|z_k)$ is the posterior probability, the *decoding* estimate of hand state x_k from neuronal firing z_k that we seek to find. $\kappa = 1/p(z_k)$, and will end up normalizing the total probability on the right to ensure that it sums to 1. $p(z_k|x_k)$ is the *likelihood*—it is estimated from the actual data used to determine encoding such as in figure 9.12. $p(x_{k-1}|z_{k-1})$ is the previous posterior probability at time index $k-1$, used to establish the initial conditions for the new *temporal prior*, which is $p(x_k|x_{k-1})$. Integrating $p(x_k|x_{k-1})p(x_{k-1}|z_{k-1})$ is the same as integrating the equations of motion for this dynamical system. It propagates the hand state at time $k-1$ to the state at time k, $p(x_k)$. If the likelihood and temporal prior are both unimodal Gaussian functions, then we can use the original Kalman filter to solve this problem recursively.

The likelihood model is $\mathrm{z}_k = \mathrm{H}_k x_k + r_k$, where H is a linear observation function matrix, and r_k is drawn from a Gaussian probability distribution. The temporal prior is $\mathrm{x}_{k+1} = \mathrm{A}_k x_k + q_k$, where A is a linear matrix of the equations of motion, and q_k is drawn from a Gaussian probability distribution.[7] Wu and colleagues [WGB+06] are careful to include a full covariance matrix, R, which is $C\mathrm{x}C$ in their noise structure for the C neurons. This is important. One knows immediately from results such as those shown in figure 9.8 that there was much interdependency and redundancy in representation of information among these neurons. The full covariance structure of R represents this covariation among each individual neuron, and allows for dependency. This is important to improving the accuracy of their results.

Using the Kalman filter equations from chapter 2, the central recursion is

$$\hat{x}_k = x_k^- + K_k(z_k - H\hat{x}_k^-)$$

The observations z_k are compared with the estimated observations $H\hat{x}_k^-$ and their difference is weighted by the Kalman gain K_k.

7. If the reader is using this discussion to decode Wu et al. [WGB+06], I apologize for switching notation to maintain consistency with earlier chapters in this book.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

The likelihood model, $z_k = H_k x_k + r_k$, is their model of the *neural code*. H is the brain. Is this a good physical representation of the dynamics of the brain? You do not need to know much about the brain other than how to get into it to write this down. This is an empirical fit of a statistical model. But the counter-question would be: Would all of your knowledge of the brain gathered over the past century enable you to do a better job in decoding its signals? Perhaps not now. But at some point, our increasing understanding of neuronal dynamics should enable us to consider whether we can do better than the present state of the field, which uses *black box* statistical models as described here.

It takes time for sensory input to travel through our nervous system to be perceived, and similarly, it takes time for motor intention to travel through our nervous system and muscle actuators to result in hand movement. Wu and colleagues [WGB+06] explored a range of time lags, l, for neuronal firing data:

$$\mathbf{z}_k = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_C \end{bmatrix} \rightarrow \pm \{l_j\}$$

and found that an optimal time lag was about 140 to 150 msec (similar to the optimal lags found by Moran and Schwartz [MS99b]).

The results of this decoding are shown in figure 9.13. Are these reconstructions good? Again we see the observation, that "we observed decoding results with relatively high correlation coefficients that were sometimes far from the true 2D hand trajectory" [WGB+06].[8]

And all of this makes a lot of sense. Again, the model is fit into a space much smaller than the brain operates within. All of the model inadequacy is being lumped into the random R and Q distributions from chapter 2. The considerations from chapter 8, in terms of mapping and transforming model results back to the space that nature operates within, seem very pertinent here.

Adding kinematic derivatives beyond one or two did not help (beyond acceleration for pinball tasks, and beyond velocity for pursuit tracking). This is not to say that such variables are not present in the kinematics of the hand. But increasing the complexity of these models comes at a price—one needs to estimate more state variables. There is a trade-off between model complexity and filter accuracy. This is very advantageous for us. If the best filter required a full computational model of the brain running on a high-density computer cluster

8. One can try to gain additional accuracy in such calculations by using *target conditioning*—essentially adding an additional boundary condition to the recursive initial value problem that the Kalman filter solves [KP08].

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

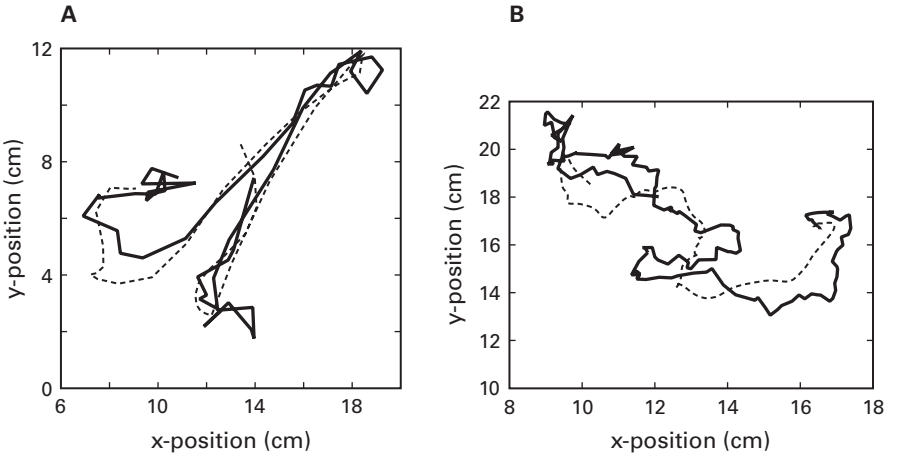


Figure 9.13
True (dashed) and reconstructed (solid) trajectories for pinball task (A) and pursuit (B) Reproduced from [WGB+06] with permission.

of the size of the universe, it would be somewhat impractical to consider using these in the laboratory or implanting them into a person.

Last, Wu and colleagues [WGB+06] compared their results with that of their previously used linear filters and polulation vectors. Their performance with Kalman filtering was better than that when using a linear filter with no explicity generative model of hand kinematics, similar to the findings of [BFT+98]. Their results were particularly better than those from the older population vector encoding. But don't discount such simpler models yet.

9.7 When Bad Models Happen to Good Monkeys

In 2002, Dawn Taylor and colleagues [TTS02] did a remarkable experiment. They used a version of population vector encoding in a three-dimensional center-out task. But they let the monkeys have visual feedback for some of these experiments, allowing their brains to do the work with their hands restrained. They compared these results with those when the monkeys used their hands. And they did one more thing—they retuned their algorithms in case the monkeys changed or adapted their neurons' tuning properties over the period of training.

Figure 9.14 demonstrates the main result. In this *coadaptive* paradigm, where the brain and the algorithm adapt to each other, there is a very significant improvement in accuracy of this neural prosthetic from day to day. This accuracy was linked to an improvement in the neurons' cosine tuning properties for direction for subjects that had visual feedback.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

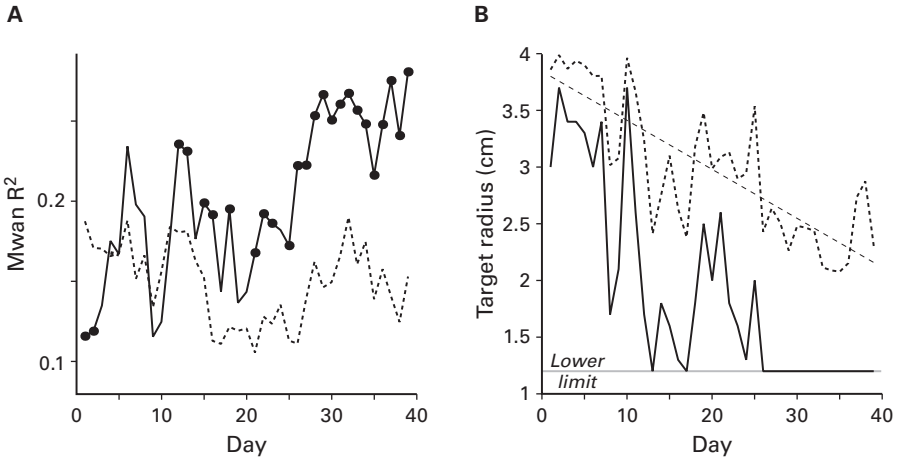


Figure 9.14
A shows the quality of cosine tuning expressed as the average R^2 (coefficient of correlation) statistic for the neurons recorded. Dotted lines are for hand-controlled subjects; solid lines are for brain-control subjects. Significant differences are indicated by solid circle markers. B shows improvement in brain-controlled accuracy by contrasting target size required for 70% accuracy (dotted line is mean; solid line is minimum). Reproduced from [TTS02] with permission.

The coadaptive brain-controlled subjects could achieve accuracy similar to hand-controlled subjects.

The brain can take a mediocre algorithm, and adjust its own tuning curves to adapt to that algorithm and produce better performance. Filters can optimize to the brain activity and motor performance. But *brains can adapt to filters*. Perhaps that is why much of the studies described in this chapter worked well. You just have to give the brain something it can work with. Hammers are not optimized for driving nails, but many of us can become skilled in using them given a chance.

Perhaps the brain has its Kalman filter adapting to our Kalman filters in these experiments. One can immediately see that we should be able to create adaptive controllers that are not at all well suited for the brain. Which types of adaptation are best for the brain's native adaptation capabilities remains a fully open question.

Similarly, this chapter has focused on extracting signals from the motor cortex. It is not at all clear that this is necessary or best. But at the time of this writing, exploring motor control prosthetics from brain signals outside of the motor cortex has not yet been explored.

Much of this book has focused on the nonlinear ensemble Kalman filters. There was an effort to apply such an ensemble filter to brain machine interfaces in 2004 [BRK04]. Brockwell and colleagues recognized the power and flexibility of such methods in that the linear Gaussian models of prior and likelihood models were no longer restrictions. But

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

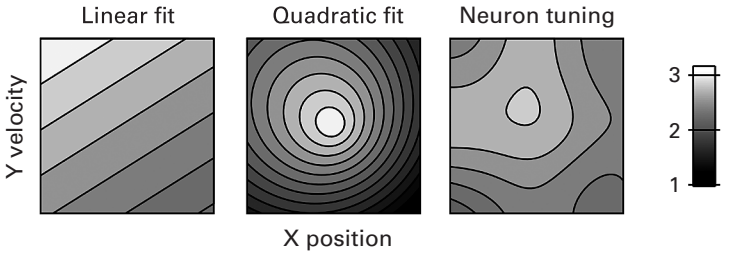


Figure 9.15
Fitting neuronal activity as a function of hand position with linear versus quadratic functions. Brightness is proportional to firing rate. Reproduced from [LOH+09] with permission.

despite the efficiency of their *particle filter*, running thousands of points as initial conditions for such an ensemble filter was not efficient enough to run in real time.[9]

More recently, Li and colleagues employed a sparse ensemble filter—the unscented Kalman filter—in a multineuron brain-machine interface [LOH+09]. There were three important advances in this work. First, they finally got beyond the linear tuning curves, as shown in figure 9.12. Figure 9.15 shows the improvement in fit for neuronal tuning curves for things such as hand position when using quadratic versus linear functions. Second, they used a ten-step autoregressive model for hand kinematics, and although they implemented this filter causally, they used an acausal model within the filter. Third, using the unscented Kalman filter, they could demonstrate real-time performance in both center-out and pursuit tasks. Comparing the performance accuracy with the linear Kalman filter from [WGB+06], they achieved improved accuracy with their tenth-order uscented Kalman filter.

9.8 Toward the Future

In spite of all the impressive developments outlined in this chapter, where is the brain? In between the neuronal output spikes and limb movement for instance, there is a void where our neurophysiology remains uninformative (other than spike tuning curves). This work has often been focused on a decoding paradigm: Neuronal firing is conditioned on arm movement, for instance, and arm movement is conditioned on neuronal firing. If one is building a brain-machine interface, recent work toward a general model that incorporates a hybrid composite of discrete and continuous states, as well as neuronal firing, is likely the most advanced framework currently available [SB07].

9. Nevertheless, computer processor and multicore architecture offer progressively improving computational capabilities, and robust particle filtering methods become more practical for real time use with each passing day [BKS07]. As a general filtering design principle, one should consider increasing the ensemble size to the limit that computational overhead permits.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

It is tempting to speculate whether we can improve brain-machine interfaces by building models not on fitting data to empirical tuning curves, but rather by building models based on neurophysiology. Such models would incorporate the properties of individual neurons, and their network connectivity, that have have been observed through electrode recordings. Such models would be fused with data—*data assimilation* [Kal03].

We remain far from such a goal at present. But our modeling sophistication, our means for making measurements within the brain, and our control theoretical tools are ready when we are ready.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026

Downloaded from direct.mit.edu/books/book/chapter-pdf/224783/9780262312080_caq.pdf by Eindhoven University of Technology user on 06 November 2026